



Computer Vision

L06: Feature Detection and Matching (Corner Detection)

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Outline

Feature Detectors

- Harris Corner Detector

- Mathematics of Harris Detector

- Other Versions of Harris Detectors

Feature Descriptors

Feature Matching

Feature Tracking



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Feature Descriptors

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Feature Detectors

- ▶ What are **good features to track**?
- ▶ **Textureless patches** are nearly **impossible to localize**.
- ▶ Patches with **large contrast changes** (gradients) are **easier to localize**.
- ▶ Straight **line segments** at a single orientation suffer from the **aperture problem**.

Measuring the Performance of the Feature Detector

- ▶ **Measuring repeatability:** Frequency with which keypoints detected in one image are found within ϵ (say, $\epsilon=1.5$) pixels of the corresponding location in a transformed image.
- ▶ **Scale invariance:** By performing the same operations at multiple resolutions in a pyramid and then matching features at the same level.
- ▶ **Rotational invariance and orientation estimation.**
- ▶ **Affine invariance:** They also respond consistently across affine deformations such as (local) perspective foreshortening.

The Simplest Possible Matching Criterion

- ▶ Weighted summed square difference:

$$E_{WSSD}(u) = \sum_i w(x_i)[I_1(x_i + u) - I_o(x_i)]^2$$

- ▶ We can only compute how stable this metric is with respect to small variations in position Δu by comparing an image patch against itself, which is known as an **auto-correlation function or surface**: $E_{AC}(\Delta u) = \sum_i w(x_i)[I_1(x_i + u) - I_o(x_i)]^2$

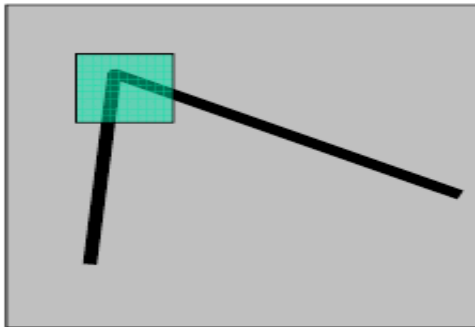


Possible Approaches to Corner Detection

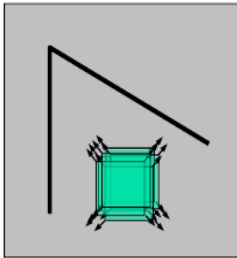
- ▶ Based on brightness of images
 - ▶ Usually image derivatives
- ▶ Based on boundary extraction
 - ▶ First step edge detection
 - ▶ Curvature analysis of edges

Harris Corner Detector

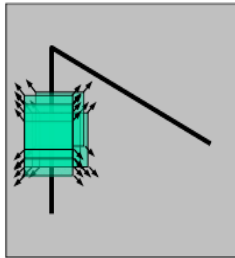
- ▶ Corner point can be recognized in a window.
- ▶ Shifting a window in any direction should give a large change in intensity.



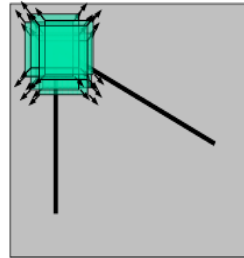
Basic Idea



“flat” region:
no change in
all directions

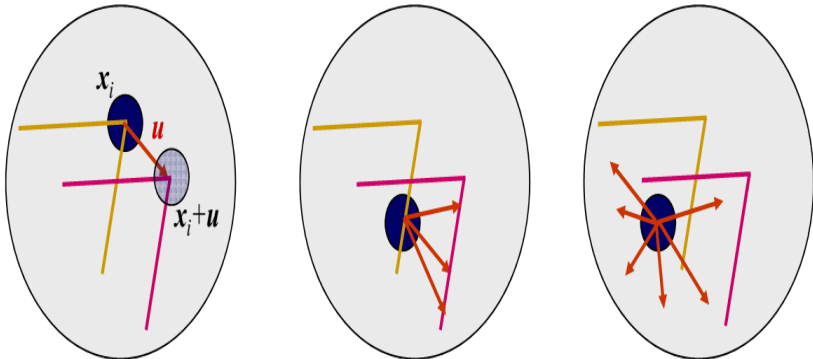


“edge”:
no change along
the edge direction



“corner”:
significant change
in all directions

Aperture Problem



1. Correlation

$$f \otimes h = \sum_k \sum_l f(k, l) h(i + k, j + l)$$

f = Image

h = Kernel

f

f_1	f_2	f_3
f_4	f_5	f_6
f_7	f_8	f_9

$\otimes$

h

h_1	h_2	h_3
h_4	h_5	h_6
h_7	h_8	h_9

$\rightarrow$

$f * h = f_1 h_1 + f_2 h_2 + f_3 h_3$
 $+ f_4 h_4 + f_5 h_5 + f_6 h_6$
 $+ f_7 h_7 + f_8 h_8 + f_9 h_9$

1. Correlation (cont'd)

$$f \otimes h = \sum_k \sum_l f(k, l) h(i + k, j + l)$$

Cross correlation

$$f \otimes f = \sum_k \sum_l f(k, l) f(i + k, j + l)$$

Auto correlation

2. Correlation Vs SSD

$$\underset{\text{minimize}}{SSD} = \sum_k \sum_l (f(k, l) - h(i + k, j + l))^2 \quad \text{Sum of Squares Difference}$$

$$\underset{\text{minimize}}{SSD} = \sum_k \sum_l (f(k, l)^2 - 2h(i + k, j + l)f(k, l) + h(i + k, j + l)^2)$$

$$SSD = \sum_k \sum_l (-2h(i + k, j + l)f(k, l))$$

$$\underset{\text{maximize}}{SSD} = \sum_k \sum_l (2h(i + k, j + l)f(k, l))$$

$$\underset{\text{maximize}}{\text{Correlation}} = \sum_k \sum_l (h(i + k, j + l)f(k, l))$$

$$f \otimes f = \sum_k \sum_l f(k, l)f(i + k, j + l)$$

3. Taylor Series

$f(x)$ Can be represented at point a in terms of its derivatives

$$f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f^{(3)}(a)}{3!}(x-a)^3 + \dots$$

4. Eigen Vectors and Eigen Values

The eigen vector, x , of a matrix A is a special vector, with the following property

$$Ax = \lambda x$$

Where λ is called eigen value

To find eigen values of a matrix A first find the roots of:

$$\det(A - \lambda I) = 0$$

Then solve the following linear system for each eigen value to find corresponding eigen vector

$$(A - \lambda I)x = 0$$

Example

$$A = \begin{bmatrix} -1 & 2 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 7 \end{bmatrix}$$

Eigen Values

$$\lambda_1 = 7, \lambda_2 = 3, \lambda_3 = -1$$

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 4 \\ 4 \end{bmatrix}, \mathbf{x}_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \mathbf{x}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Eigen Vectors

Eigen Values

$$\det(A - \lambda I) = 0$$

$$\det\left(\begin{bmatrix} -1 & 2 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 7 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\right) = 0$$

$$\det\left(\begin{bmatrix} -1-\lambda & 2 & 0 \\ 0 & 3-\lambda & 4 \\ 0 & 0 & 7-\lambda \end{bmatrix}\right) = 0$$

$$(-1-\lambda)((3-\lambda)(7-\lambda)-0) = 0$$

$$(-1-\lambda)(3-\lambda)(7-\lambda) = 0$$

$$\lambda = -1, \quad \lambda = 3, \quad \lambda = 7$$

Eigen Vectors

$$\lambda = -1$$

$$(A - \lambda I)x = 0$$

$$\left(\begin{bmatrix} -1 & 2 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 7 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 2 & 0 \\ 0 & 4 & 4 \\ 0 & 0 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$0 + 2x_2 + 0 = 0$$

$$0 + 4x_2 + 4x_3 = 0$$

$$0 + 0 + 8x_3 = 0$$

$$x_1 = 1, \quad x_2 = 0, \quad x_3 = 0$$

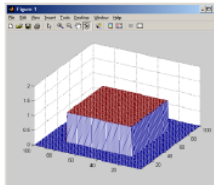
Mathematics of Harris Detector

- Change of intensity for the shift (u,v)

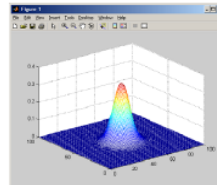
$$E(u,v) = \sum_{x,y} \left[\underbrace{I(x+u, y+v)}_{\text{shifted intensity}} - \underbrace{I(x,y)}_{\text{intensity}} \right]^2$$

Auto-correlation

Window functions →



UNIFORM



GAUSSIAN

Mathematics of Harris Detector (cont'd)

$$E(u, v) = \sum_{x, y} \underbrace{w(x, y)}_{\text{window function}} \underbrace{[I(x + u, y + v) - I(x, y)]}_{\text{shifted intensity} - \text{intensity}}^2$$

$$E(u, v) = \sum_{x, y} \underbrace{w(x, y)}_{\text{window function}} \underbrace{[I(x, y) + uI_x + vI_y - I(x, y)]}_{\text{shifted intensity} - \text{intensity}}^2 \quad \text{Taylor Series}$$

$$E(u, v) = \sum_{x, y} w(x, y) [uI_x + vI_y]^2$$

$$E(u, v) = \sum_{x, y} w(x, y) \left[\begin{pmatrix} u & v \end{pmatrix} \begin{pmatrix} I_x \\ I_y \end{pmatrix} \right]^2$$

$$E(u, v) = \sum_{x, y} w(x, y) \begin{pmatrix} u & v \end{pmatrix} \begin{pmatrix} I_x \\ I_y \end{pmatrix} \begin{pmatrix} I_x & I_y \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}$$

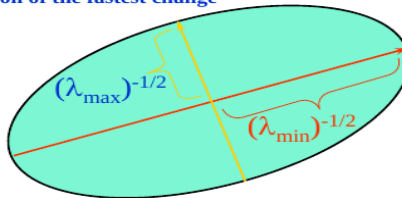
$$E(u, v) = \begin{pmatrix} u & v \end{pmatrix} \left[\sum_{x, y} w(x, y) \begin{pmatrix} I_x \\ I_y \end{pmatrix} \begin{pmatrix} I_x & I_y \end{pmatrix} \right] \begin{pmatrix} u \\ v \end{pmatrix} \quad E(u, v) = \begin{pmatrix} u & v \end{pmatrix} M \begin{pmatrix} u \\ v \end{pmatrix}$$

Mathematics of Harris Detector (cont'd)

$$E(u, v) = \begin{pmatrix} u & v \end{pmatrix} M \begin{pmatrix} u \\ v \end{pmatrix} \quad M = \sum_{x,y} w(x, y) \begin{bmatrix} I_x I_x & I_x I_y \\ I_x I_y & I_y I_y \end{bmatrix}$$

- $E(u, v)$ is an equation of an ellipse, where M is the covariance
- Let λ_1 and λ_2 be eigenvalues of M

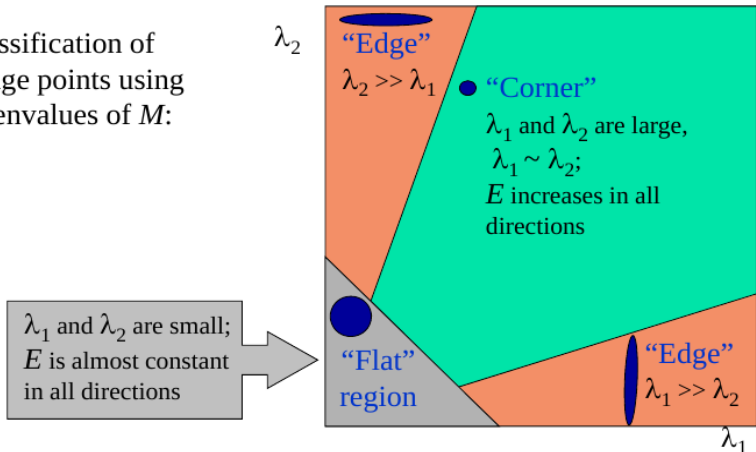
direction of the fastest change



direction of the slowest change

Mathematics of Harris Detector (cont'd)

Classification of image points using eigenvalues of M :



Mathematics of Harris Detector (cont'd)

- Measure of cornerness in terms of λ_1, λ_2

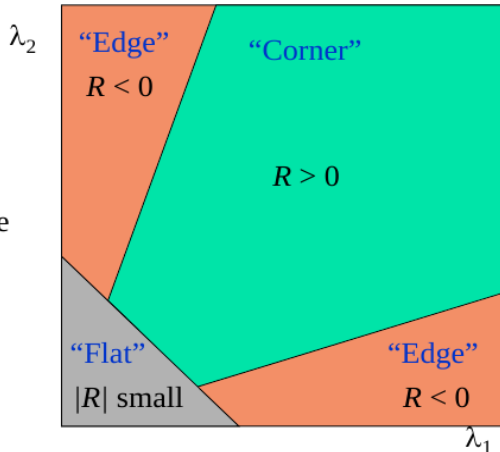
$$M = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

$$R = \det M - k(\text{trace}M)^2$$

$$R = \lambda_1 \lambda_2 - k(\lambda_1 + \lambda_2)^2$$

Mathematics of Harris Detector (cont'd)

- R depends only on eigenvalues of M
- R is large for a **corner**
- R is negative with large magnitude for an **edge**
- $|R|$ is small for a **flat** region



Outline of a Basic Feature Detection Algorithm

1. Compute the horizontal and vertical derivatives of the image I_x and I_y by convolving the original image with derivatives of Gaussians (Section 3.2.3).
2. Compute the three images corresponding to the outer products of these gradients. (The matrix A is symmetric, so only three entries are needed.)
3. Convolve each of these images with a larger Gaussian.
4. Compute a scalar interest measure using one of the formulas discussed above.
5. Find local maxima above a certain threshold and report them as detected feature point locations.

Other Versions of Harris Detectors

$$R = \lambda_1 - \alpha \lambda_2$$

Triggs

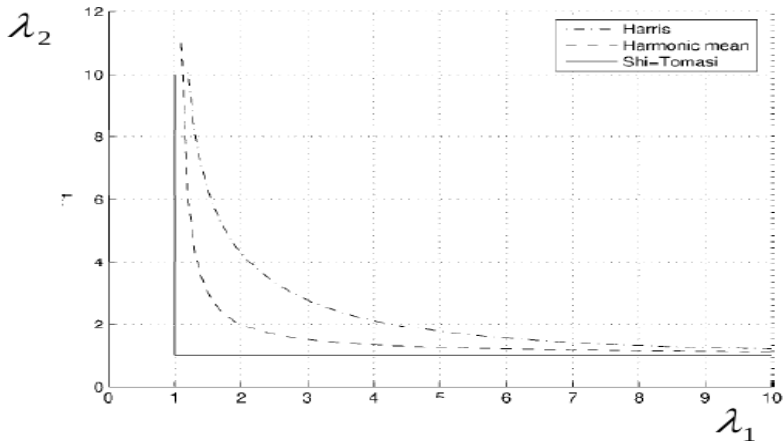
$$R = \frac{\det(M)}{\text{trace}(M)_1} = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$$

Szeliski (Harmonic mean)

$$R = \lambda_1$$

Shi-Tomasi

Other Versions of Harris Detectors (cont'd)





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Feature Matching

Feature Tracking

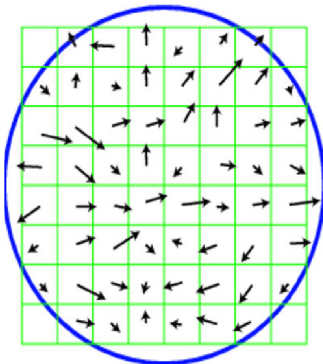
Feature Descriptors

- ▶ The local motion around each feature point may be **mostly translational**.
 - ▶ In this case, simple error metrics, such as the **sum of squared differences** or **normalized cross-correlation** can be used to directly compare the intensities in small patches around each feature point.
- ▶ The **local appearance of features** will change in **orientation, scale**, and undergo **affine deformations**.
- ▶ **How can we make image descriptors more invariant to such changes?**

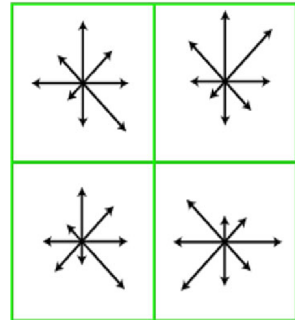
Multi-scale Oriented Patches (MOPS) Detector

- ▶ To compensate for **slight inaccuracies** in the feature point detector (**location**, **orientation**, and **scale**), MOPS are **sampled at a spacing of five pixels** relative to the detection scale.
- ▶ Using a **coarser level of the image pyramid** to avoid **aliasing**.
- ▶ To **compensate for affine photometric variations** (**linear exposure changes** or **bias and gain**), **patch intensities are re-scaled** so that their **mean is zero** and **variance is one**.

Scale Invariant Feature Transform (SIFT) Detector



(a) image gradients



(b) keypoint descriptor



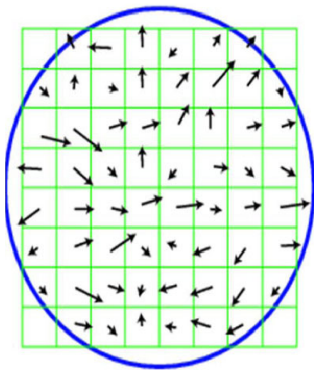
PCA-SIFT Detector

- ▶ It computes the **x and y (gradient) derivatives** over a **39x39 patch** and then reduces the resulting **3042-dimensional vector** to **36** using **principal component analysis (PCA)**.
- ▶ **SURF** uses box filters to **approximate the derivatives and integrals used in SIFT**.

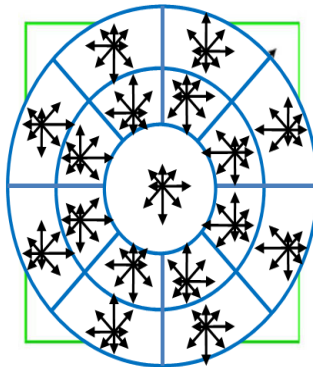
Gradient Location-orientation Histogram (GLOH) Detector

- ▶ It is a **variant on SIFT** that uses a **log-polar binning structure instead of the four quadrants**.
- ▶ The **spatial bins** are of **radius 6, 11, and 15**, with **eight angular bins** (except for the central region), for a total of **17 spatial bins and 16 orientation bins**.
- ▶ The **272-dimensional histogram** is then projected onto a **128-dimensional descriptor using PCA** trained on a large database.
- ▶ **It is found that GLOH, which has the best performance overall, outperforms SIFT by a small margin.**

GLOH



(a) image gradients



(b) keypoint descriptor



Steerable Filters

- ▶ They are combinations of **derivative of Gaussian filters** that permit the **rapid computation of even and odd** (symmetric and anti-symmetric) **edge-like and corner-like features** at all possible orientations.
- ▶ They too are somewhat **insensitive to localization and orientation errors**.



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- Feature Detectors
- Feature Descriptors
- Feature Matching**
- Feature Tracking

Feature Matching Components

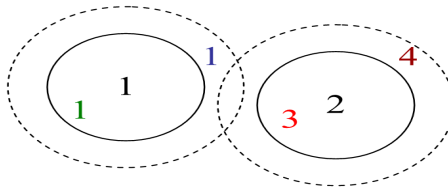
1. Selecting a matching strategy that **determines which correspondences are passed** on to the next stage for further processing.
2. Devising efficient data structures and algorithms to perform this matching as quickly as possible.



Notes

- ▶ **Determining which feature matches** are **reasonable to process** further **depends on the context** in which the matching is being performed.
- ▶ Some **may not match** because they are **occluded** or **their appearance has changed too much**.

False Positives and Negatives



The black digits 1 and 2 are features being matched against a database of features in other images. At the current threshold setting (the solid circles), the green 1 is a true positive (good match), the blue 1 is a false negative (failure to match), and the red 3 is a false positive (incorrect match). If we set the threshold higher (the dashed circles), the blue 1 becomes a true positive but the brown 4 becomes an additional false positive.



False Positives and Negatives (cont'd)

- ▶ **TP: true positives**, i.e., number of correct matches.
- ▶ **FN: false negatives**, matches that were not correctly detected.
- ▶ **FP: false positives**, proposed matches that are incorrect.
- ▶ **TN: true negatives**, non-matches that were correctly rejected.

Confusion Matrix (Contingency Table)

	True matches	True non-matches	
Predicted matches	TP = 18	FP = 4	P' = 22
Predicted non-matches	FN = 2	TN = 76	N' = 78
	P = 20	N = 80	Total = 100

PPV = 0.82

TPR = 0.90	FPR = 0.05
------------	------------

ACC = 0.94

- ▶ The columns sum up to the actual number of positives (P) and negatives (N).
- ▶ The rows sum up to the predicted number of positives (P') and negatives (N').
- ▶ True positive rate (TPR), false positive rate (FPR), positive predictive value (PPV), the accuracy (ACC).

Converting to Unit Rate

- true positive rate (**TPR**),

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{\text{TP}}{P};$$

- false positive rate (**FPR**),

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} = \frac{\text{FP}}{N};$$

- positive predictive value (PPV),

$$\text{PPV} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{\text{TP}}{P'};$$

- accuracy (ACC),

$$\text{ACC} = \frac{\text{TP} + \text{TN}}{P + N}.$$



Note

Information/Document Retrieval

The term **precision** (how many returned documents are relevant) is used instead of **PPV**.

The term **recall** (what fraction of relevant documents was found) is used instead of **TPR**.

Efficient Matching

- ▶ A better approach is to devise an **indexing structure**, such as a **multi-dimensional search tree** or a **hash table**, to rapidly search for features near a given feature.
- ▶ One of the simpler techniques to implement is **multi-dimensional hashing**, which maps descriptors into fixed size buckets based on some function applied to each descriptor vector.
- ▶ A simple **example of hashing** is the **Haar wavelets**.



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Feature Tracking

- ▶ Find a set of likely **feature locations** in a first image and to then **search for their corresponding locations in subsequent images**.
- ▶ This kind of detect then track approach is more widely used for **video tracking applications**.
 - ▶ Where the expected **amount of motion and appearance deformation** between adjacent frames is **expected to be small**.



Feature Tracking (cont'd)

- ▶ **Normalized Cross-correlation**: if the images are undergoing brightness change.
- ▶ **Hierarchical search strategy**: if the search range is large.

References

- ▶ Richard Szeliski, Computer Vision: Algorithms and Applications, 2nd Ed., Springer-Verlag London, 2020.
- ▶ Reinhard Klette, Concise Computer Vision: An Introduction into Theory and Algorithms, Springer-Verlag London 2014.
- ▶ Mubarak Shah CV Course: **Lecture 4**
(https://www.youtube.com/watch?v=S4mMhuLHrsQ&list=PLd3hISJsX_ImKP68wfKZJV IPTd8le5u-9&index=4).